Chapter 6 Intro to Networking Exam

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Score: 93% Passing Score: 80%

Question 1: ✔ Correct

You manage a local area network with several switches. A new employee has started today, so you connect her workstation to a switch port.

After connecting the workstation, you find that the workstation cannot get an IP address from the DHCP server. You check the link and status lights and see that the connection is working properly. A ping to the loopback address on the workstation succeeds. No other computers seem to have the problem.

Which of the following is the most likely cause of the problem?

- Half-duplex setting on the switch and workstation
- Incorrect default gateway setting
- ➡ Incorrect VLAN assignment
- Switching loop

EXPLANATION

The most likely cause is that the switch port is a member of a VLAN that is different from the VLAN for the DHCP server and other devices. It is possible that unused ports on the switch were assigned to a VLAN that is different from the VLAN used by other devices.

The duplex setting would probably not prevent traffic between the workstation and the switch; it would simply mean that both devices would perform collision detection. A problem might occur if one device were manually configured for full-duplex, and the other were configured for half-duplex.

A switching loop occurs when there are multiple active paths between two switches. Switching loops lead to incorrect entries in a MAC address table, making a device appear to be connected to the wrong port and causing unicast traffic being circulated in a loop between switches. Switching loops would typically affect multiple devices, not just one.

The default gateway setting affects whether the device can communicate with hosts on different subnets, but this value is typically received from the DHCP server.

REFERENCES

LabSim for Network Pro, Section 6.7.

Question 2: ✔ Correct

You run a small network for your business that has a single router connected to the internet and a single switch. You keep sensitive documents on a computer that you would like to keep isolated from other computers on the network. Other hosts on the network should not be able to communicate with this computer through the switch, but you still need to access the network through the computer.

Which of the following should you use in this situation?

- ➡ VLAN
- Port security
- Spanning tree
- VPN

EXPLANATION

You should define virtual LANs (VLANs) on the switch. With a VLAN, a port on the switch is associated with a VLAN. Only devices connected to ports that are members of the same VLAN can communicate with each other. Routers are used to allow communication between VLANs if necessary.

Use a virtual private network (VPN) to connect two hosts securely through an unsecured network (such as the internet). VPN tunneling protocols protect data as it travels through the unsecured network. Spanning tree is a switch feature that allows for redundant paths between switches. Port security is a method of requiring authentication before a network connection is allowed.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 3: ✔ Correct

Match each type of switch on the left with its corresponding characteristics on the right. Each switch type may be used once, more than once, or not at all.

Commonly sold at retail stores.

✔ Unmanaged switch

Provides port security features.

✔ Managed switch

Supports VLANs.

✔ Managed switch

Provides very few configuration options.

✔ Unmanaged switch

Can be configured over a network connection.

✔ Managed switch

Can be configured over a dedicated communication channel.

✔ Managed switch

EXPLANATION

The low-end switches available from many retail stores cannot be configured. These are called unmanaged switches. To implement an unmanaged switch, you simply plug it in to a power outlet and connect your network devices with UTP cables. While unmanaged switches are convenient and easy to implement, they lack many of the advanced management and security features available on managed switches. For example, managed switches provide port security and support VLANs.

REFERENCES

LabSim for Network Pro, Section 6.1.

Question 4: ✔ Correct

When you configure VLANs on a switch, which of the following is used to identify a device's VLAN membership?

- ➡ Switch port
- MAC address
- IP address
- Host name

EXPLANATION

VLAN membership is configured by assigning a switch port to a VLAN. A switch can have multiple VLANs configured on it, but each switch port can only be a member of a single VLAN. All devices connected to a switch port are members of the same VLAN.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 5: — Partially Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You did not complete the lab correctly.

View Lab Report

Question 6: ✔ Correct

Which of the following statements describe how VLANs affect broadcast traffic within an internetwork? (Select two.)

- [ ] Devices on separate VLANs share the same subnet address.
- [x] Broadcast traffic is transmitted only within a VLAN.
- [ ] Broadcast traffic is transmitted to all devices on all VLANs.
- [ ] Broadcast traffic is only transmitted on VLAN1.
- [x] Devices on the same VLAN have the same subnet address.

EXPLANATION

VLANs allow computers to be grouped into a common broadcast domain regardless of their physical location on the network. Broadcast traffic is seen only by computers belonging to the same VLAN. Devices sharing a VLAN ID must also share the same subnet address so that traffic can be routed between VLANs.

Broadcast traffic is only transmitted to devices belonging to the same VLAN as the device that sent the broadcast. Broadcast traffic is not forwarded to the other VLANs.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 7: ✔ Correct

You can create a virtual LAN using which of the following?

- Hub
- Gateway
- Router
- ➡ Switch

EXPLANATION

Use a switch to create virtual LANs (VLANs). The various ports on a switch can be assigned to a specific VLAN to create logically distinct networks on the same physical network topology.

Routers, gateways, and hubs are common network devices, but they do not support the creation of VLANs.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 8: ✔ Correct

You manage a single subnet with three switches. The switches are connected to provide redundant paths between the switches.

Which feature allows the switches to pass VLAN traffic between the switches?

- Bonding
- PoE
- 802.1x
- Spanning tree
- ➡ Trunking

EXPLANATION

A *trunk* port is used to connect two switches together. A trunk port is a member of all VLANs and carries traffic between the switches. When trunking is used, frames that are sent over a trunk port are tagged by the first switch with the VLAN ID so that the receiving switch knows which VLAN the frame belongs to. The trunking protocol describes the format that switches use for tagging frames with the VLAN ID. Because end devices do not understand the VLAN tags, the tag is removed from the frame by the switch before the frame is forwarded to the destination device.

802.1x is an authentication protocol used with port security (or port authentication). Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Power over Ethernet (PoE) supplies power to end devices through the RJ45 Ethernet switch port. Bonding groups multiple switch ports into a single logical port, allowing both ports to be used to communicate with a single device.

REFERENCES

LabSim for Network Pro, Section 6.5.

Question 9: ✔ Correct

Which of the following benefits apply only to creating VLANs with switches and not to segmenting the network with regular switches?

- ➡ You can create multiple broadcast domains.
- You can segment the network based on physical location.
- You can create multiple collision domains.
- You can provide guaranteed bandwidth between two devices.

EXPLANATION

By creating a VLAN, you create multiple broadcast domains, with each VLAN being in its own broadcast domain. Using switches without VLANs lets you segment the network based on physical location, create multiple collision domains, and provide guaranteed bandwidth between devices.

REFERENCES

LabSim for Network Pro, Section 6.5.

Question 10: ✔ Correct

A switch running STP is in the listening state. A message destined for a different network segment arrives at the switch.

Which of the following best describes what the switch will do?

- It does not record address or segment information. However, the message is forwarded.
- It places the source address and segment in its database and forwards the message.
- ➡ It does not record address or segment information. It does not forward the message.
- It places the source address and segment in its database, but does not forward the message.

EXPLANATION

A switch in listening state receives packets, but does not build its database or forward them. Switches in the listening state are in the process of defining their role on the network.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 11: ✔ Correct

Which statements accurately describe the port states of both bridges and switches? (Select two.)

- [x] Ports in a blocked state still receive BPDUs.
- [ ] Ports in a blocked state cannot receive BPDUs.
- [x] In the learning state, the MAC address table can be populated, but frames are not forwarded.
- [ ] In the learning state, the MAC address table cannot be populated because the port is blocked.
- [ ] In the learning state, all ports are in a forwarding state.

EXPLANATION

For both bridges and switches:

- In the learning state, ports do not forward frames, but still populate the MAC address table based on frames received.
- In the blocking state, ports receive BPDUs, but do not forward frames.
- In the listening state, all ports are blocked.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 12: ✔ Correct

You run a small network for your business that has a single router connected to the internet and a single switch. You keep sensitive documents on a computer that you would like to keep isolated from other computers on the network. Other hosts on the network should not be able to communicate with this computer through the switch, but you still need to access the network through the computer.

What should you use for this situation?

- ➡ VLAN
- Port security
- VPN
- Spanning tree

EXPLANATION

Define virtual LANs (VLANs) on the switch. With a VLAN, a port on the switch is associated with a VLAN. Only devices connected to ports that are members of the same VLAN can communicate with each other. Routers are used to allow communication between VLANs if necessary.

Use virtual private network (VPN) to connect two hosts securely through an unsecured network (such as the internet). VPN tunneling protocols protect data as it travels through the unsecured network. Spanning tree is a switch feature that allows redundant paths between switches. Port security is a method of requiring authentication before a network connection is allowed.

REFERENCES

LabSim for Network Pro, Section 6.7.

Question 13: ✔ Correct

This question includes an image to help you answer the question. View Image

Computers A and B are on the same VLAN and are separated by two switches as shown in the exhibit. Computer A sends a frame to Computer B.

Which of the following best describes the composition of the frame as it travels from A to B?

- Computer A sends a normal frame. The first switch appends a VLAN ID to the frame. The VLAN ID remains on the frame through the second switch up to Computer B.
- ➡ Computer A sends a normal frame. The first switch appends a VLAN ID to the frame. The second switch removes the VLAN ID before forwarding it to Computer B.
- Computer A appends a VLAN ID to the frame. It travels from switch to switch and arrives at Computer B where it removes the VLAN ID.
- Computer A sends a normal frame. The first switch forwards the frame to the second switch, where the VLAN ID is appended to the frame and forwarded to Computer B.

EXPLANATION

VLAN IDs are understood only by switches and are used by switches for inter-switch traffic. The first switch appends the VLAN ID, and the second switch removes it.

REFERENCES

LabSim for Network Pro, Section 6.5.

Question 14: ✔ Correct

You need to configure spanning tree on a Cisco switch. You'd like to use a protocol that conforms to the 802.1w standards.

Which protocol should you use?

- ➡ Rapid PVST+
- EtherChannel
- PortFast
- BPDU guard

EXPLANATION

Rapid PVST+ is the IEEE 802.1w standard. RSTP improves convergence by actively confirming that a switch is ready to transition to a forwarding state, eliminating the listening and learning stages. RSTP defines several new types of links and uses fewer spanning tree states.

PortFast allows a port connected to a host to go into a forwarding state almost immediately. PortFast ports must be used only for connections to end user devices, such as workstations or server systems. PortFast should not be used when the port is connected to a bridge, a switch, or another STP speaking device. EtherChannel combines multiple parallel physical links into a single logical channel. The switch treats the logical channel as a single interface that provides redundancy, as well as load sharing among the links in the channel. BPDU guard is frequently used with PortFast to ensure the stability of the STP topology. BPDU guard disables a port configured by PortFast if that port receives BPDUs.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 15: ✔ Correct

Which problem does the spanning tree protocol prevent?

- Packet blocking on backup bridge ports to allow switches to forward frames to all BPDUs.
- ➡ Switching loops from developing when redundant paths are implemented between switches.
- Buffer overflows by defining source-quench messages that a receiving device sends to the transmitting device.
- Routing tables from becoming outdated by decreasing the convergence time on a network.

EXPLANATION

The spanning tree protocol is a long-standing protocol that runs in the background of bridged and switched networks to keep message loops from occurring.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 16: ✔ Correct

Match the Cisco device password type on the left with its function on the right.

Controls the ability to log on through a LAN or WAN interface configured on the device.

✔ VTY

Controls the ability to connect to the device using a web browser using HTTPS.

✔ SDM

Controls the ability to connect to the device using a direct connection.

✔ Console

EXPLANATION

The following table lists three of the most common password types that you can configure on Cisco devices, including switches and routers:

Password Type	Description
Console	Controls the ability to log on to the device through a console connection.
VTY	Controls the ability to log on to the device using a virtual terminal (VTY) connection.
EXEC mode	Controls the ability to switch to configuration modes.

REFERENCES

LabSim for Network Pro, Section 6.1.

Question 17: ✔ Correct

Switches running STP are in the process of exchanging BPDUs and redefining their roles. Which port state are the switches currently in?

- ➡ Listening
- Disabled
- Forwarding
- Blocking
- Learning

EXPLANATION

Switches that are exchanging STP configuration information to define their roles are in the listening state. After listening, designated bridges progress to learning and then forwarding. Backup bridges return to blocking.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 18: ✔ Correct

Which of the following are reasons to configure VLANs on a switch as opposed to using switches without VLANs? (Select two.)

- [x] Increased security
- [x] Increased number of broadcast domains
- [ ] Redundant paths between two hosts
- [ ] Allowing broadcast traffic between subnets
- [ ] Increased number of collision domains

EXPLANATION

Create VLANs to increase the number of broadcast domains and implement security. Each VLAN is in its own broadcast domain. Broadcast traffic within the VLAN goes only to the members of the VLAN. Members of one VLAN can only communicate with members in the same VLAN through the switch. A router or a Layer 3 switch is required to enable inter-VLAN communication.

Using switches increases the number of collision domains because each switch port is its own collision domain. Using VLANs does not offer this additional advantage. Spanning tree lets you have loop-free redundant paths through a switched network. Broadcast traffic does not travel between routers.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 19: — Partially Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You did not complete the lab correctly.

View Lab Report

Question 20: ✔ Correct

You are configuring a switch so that you can manage it using PuTTY from the same network segment as the switch. On the switch, you enter the following commands:

```
switch#config terminal
switch(config)#interface vlan 1
switch(config-if)#ip address 192.168.1.10 255.255.255.0
```

Will this configuration work?

- Yes, the switch can now be accessed by PuTTY using the IP address 192.168.1.10.
- No, the switch needs to obtain an IP address from the DHCP server using the **ip address dhcp** command.
- No, the **ip default-gateway** command needs to be set.
- ➡ No, the **no shutdown**command needs to be entered.

EXPLANATION

By default, the Vlan1 interface is set to administratively down, preventing remote access. Use the following commands to configure the switch IP address and allow management:

```
switch#config terminal
switch(config)#interface vlan 1
switch(config-if)#ip address 192.168.1.10 255.255.255.0
switch(config-if)#no shutdown
```

Because the switch is being accessed from the same network segment, the **ip default-gateway** command doesn't need to be used. The **ip address dhcp** command only allows the switch to obtain an IP address using DHCP.

REFERENCES

LabSim for Network Pro, Section 6.2.

Question 21: ✔ Correct

You manage a network with a single switch. On each switch port, a hub connects multiple devices to the switch.

Which condition are you most likely to experience on the network?

- Echo
- Attenuation
- ➡ Collisions
- Crosstalk
- Interference

EXPLANATION

A collision occurs when two devices that share the same media segment transmit at the same time. In a switched network, collisions should only occur on ports that have multiple devices attached. To eliminate collisions, connect only a single device to each switch port.

Interference is an electrical signal on a wire that is not part of the original signal sent on the wire. Common sources of interference include nearby generators, motors (such as elevator motors), radio transmitters, welders, transformers, and fluorescent lighting. Crosstalk is interference that is caused by signals within the twisted pairs of wires. Crosstalk is often introduced within connectors where the twists are removed to add the connector. Crosstalk can also occur where wires are crushed or where the plastic coating is worn.

Attenuation is the loss of signal strength from one end of a cable to the other. The longer the cable, the more attenuation. Echo occurs when some of the signal is reflected back to the transmitter, distorting the signal. Echo is caused by connecting cables and devices with different impedance ratings.

REFERENCES

LabSim for Network Pro, Section 6.7.

Question 22: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 23: ✔ Correct

You manage a network that uses a single switch. All ports within your building connect through the single switch.

In the lobby of your building are three RJ45 ports connected to the switch. You want to allow visitors to plug into these ports to gain internet access, but they should not have access to any other devices on your private network. Employees connected throughout the rest of your building should have both private and internet access.

Which feature should you implement?

- PoE
- ➡ VLANs
- Port authentication
- Spanning tree

EXPLANATION

Use VLANs to segregate hosts based on switch ports. You could define two VLANs, one for employees connected throughout the building, and another for the ports in the lobby. The ports in the lobby would have only internet access, while devices connected to ports in the rest of the building could communicate with other devices within the same VLAN.

Use port authentication to control access to the network based on things such as username and password. Port authentication would allow or deny access, but would not restrict access once authenticated or provide any type of access if not authenticated.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Power over Ethernet (PoE) supplies power to end devices through the RJ45 Ethernet switch port.

REFERENCES

LabSim for Network Pro, Section 6.7.

Question 24: ✔ Correct

You manage a network that uses a single switch. All ports within your building connect through the single switch.

In the lobby of your building are three RJ45 ports connected to the switch. You want to allow visitors to plug into these ports to gain internet access, but they should not have access to any other devices on your private network. Employees connected throughout the rest of your building should have both private and internet access.

Which feature should you implement?

- NAT
- ➡ VLANs
- DMZ
- Port authentication

EXPLANATION

Use VLANs to segregate hosts based on switch ports. You can define two VLANs, one for employees connected throughout the building, and another for the ports in the lobby. The ports in the lobby would have only internet access, while devices connected to ports in the rest of the building could communicate with other devices within the same VLAN.

Use port authentication to control access to the network based on things such as username and password. Port authentication would allow or deny access, but would not restrict access once authenticated or provide any type of access if not authenticated.

A demilitarized zone (DMZ) is a buffer network (or subnet) that sits between the private network and an untrusted network (such as the internet). Network address translation (NAT) modifies the IP addresses in packets as they travel from one network (such as a private network) to another (such as the internet). NAT allows you to connect a private network to the internet without obtaining registered addresses for every host. Hosts on the private network share the registered IP addresses.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 25: ✔ Correct

Which command would you use on a switch to enable management from a remote network?

- **no shutdown**
- ➡ **ip default-gateway 192.168.10.185**
- **ip address 192.168.10.185 255.255.255.0**
- **ip address dhcp**

EXPLANATION

To enable management from a remote network, configure the default gateway. Use the following command in global configuration mode:

switch(config)#ip default-gateway IP_address

REFERENCES

LabSim for Network Pro, Section 6.2.

Question 26: ✔ Correct

Your company is a small start-up that has leased office space in a building shared by other businesses. All businesses share a common network infrastructure. A single switch connects all devices in the building to the router that provides internet access.

You would like to make sure that your computers are isolated from computers used by other companies. Which feature should you request to have implemented?

- ➡ VLAN
- Spanning tree
- Port security
- VPN

EXPLANATION

You should define virtual LANs (VLANs) on the switch. With a VLAN, a port on the switch is associated with a VLAN. Only devices connected to ports that are members of the same VLAN can communicate with each other. Routers are used to allow communication between VLANs if necessary.

Use virtual private network (VPN) to connect two hosts securely through an unsecured network (such as the internet). VPN tunneling protocols protect data as it travels through the unsecured network. Spanning tree is a switch feature that allows for redundant paths between switches. Port security is a method of requiring authentication before a network connection is allowed.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 27: ✔ Correct

When configuring VLANs on a switch, what type of switch ports are members of all VLANs defined on the switch?

- Any port not assigned to a VLAN.
- ➡ Trunk ports.
- Uplink ports.
- Each port can only be a member of a single VLAN.
- Gigabit and higher Ethernet ports.

EXPLANATION

A trunk port is a member of all VLANs defined on a switch and carries traffic between the switches. When trunking is used, frames that are sent over a trunk port are tagged by the first switch with the VLAN ID so that the receiving switch knows which VLAN the frame belongs to. Typically, uplink ports (which are faster than the other switch ports) are used for trunk ports, although any port can be designated as a trunking port.

On an unconfigured switch, ports are members of a default VLAN (often designated VLAN 1). When you remove a port's VLAN membership, it is reassigned back to the default VLAN. Therefore, the port is always a member of one VLAN.

REFERENCES

LabSim for Network Pro, Section 6.5.

Question 28: ✕ Incorrect

You manage a network with two switches. The switches are connected together through their Gigabit Ethernet uplink ports.

You define VLAN 1 and VLAN 2 on each switch. A device on the first switch in VLAN 1 needs to communicate with a device on the same switch which is in VLAN 2.

What should you configure so that the two devices can communicate?

- PoE
- Spanning tree
- ➡ Routing
- ~~Trunking~~
- Mirroring

EXPLANATION

In a typical configuration with multiple VLANs and a single or multiple switches, workstations in one VLAN will not be able to communicate with workstations in other VLANs. To enable inter-VLAN communication, you will need to use a router (or a Layer 3 switch).

Trunking is used to configure switch ports to carry VLAN traffic between switches or between a router and a switch. If you configured a single router to connect to the switch with a single physical interface, you would have to configure trunking on that interface in addition to routing. Trunking by itself would not enable the two devices to communicate.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Mirroring sends traffic from all switch ports to a switch port you designate as the mirrored port. Power over Ethernet (PoE) supplies power to end devices through the RJ45 Ethernet switch port.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 29: ✔ Correct

What does the **ip address dhcp** command allow you to do?

- ➡ Configure the switch to obtain an IP address from a DHCP server.
- Configure the switch to act as a DHCP server.
- Specify the DHCP relay server for forwarding DHCP packets.
- Send the DHCP server address to all connected devices.

EXPLANATION

You can use the **ip address dhcp** command to configure a switch or router to get its IP address from a DHCP server. The DHCP server can be configured to deliver the default gateway and DNS server addresses to the Cisco device as well. A manually configured default gateway address overrides any address received from the DHCP server.

REFERENCES

LabSim for Network Pro, Section 6.2.

Question 30: ✔ Correct

You want to reduce collisions by creating separate collision domains and virtual LANs. Which of the following devices should you choose?

- Router
- ➡ Switch
- Bridge
- Active hub

EXPLANATION

Use a switch to create additional collision domains on a LAN. A switch filters an entire network and creates virtual LANs inside it rather than dividing it into separate internetworks as a router does.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 31: ✔ Correct

The FastEthernet 0/0 interface on a switch is currently disabled. You need to enable it so a workstation can be connected to it.

Drag the command on the left to the appropriate configuration step on the right. It is possible that not all commands are required.

Enter global configuration mode.Enter interface configuration mode.Enable the interface.

✔ `conf t`

✔ `int fa0/0`

✔ `no shut`

Verify that the interface is enabled.

✔ `show interface status`

EXPLANATION

To complete the requirements of this scenario, you need to use the following commands:

- Enter global configuration mode: **conf t**
- Enter interface configuration mode: **int fa0/0**
- Enable the interface: **no shut**
- Verify the interface status: **show interface status**

REFERENCES

LabSim for Network Pro, Section 6.3.

Question 32: ✔ Correct

This question includes an image to help you answer the question. **View Image**

You have just connected four switches as shown in the Exhibit.

Assuming the default switch configuration, how can you force switch C to become the root bridge?

- Remove link cable 6 from the configuration.
- Remove link cable 1 from the configuration.
- Configure a priority number of 61440 for switch C.
- ➡ Configure a priority number of 4096 for switch C.
- Remove link cables 1 and 6 from the configuration.

EXPLANATION

To force a specific switch to become the root bridge, configure a priority number lower than the default (32768). The switch with the lowest bridge ID becomes the root bridge. The bridge ID is composed of two parts: a bridge priority number and the MAC address assigned to the switch. When the default priority is used for all switches, the switch with the lowest MAC address becomes the root bridge.

REFERENCES

LabSim for Network Pro, Section 6.6.

Question 33: ✔ Correct

While viewing the status of the interfaces on a Cisco switch, you see an abnormally large number of oversized Ethernet frames being received on one interface. This interface is connected to a workstation located on the second floor.

What could cause this to happen?

- ➡ The workstation's network board is jabbering.
- A strong EMI emitter is near the cable that connects the workstation to the interface.
- Two hosts on the network have the same MAC address.
- Collisions between two or more network hosts are occurring.
- An incorrect subnet mask has been configured on the workstation.

EXPLANATION

Frames that are too long are typically caused by a faulty network card that jabbers (constantly sending garbage data).

On a hub-based Ethernet network, these errors could possibly be caused by collisions. However, because a switch is being used in this scenario, collisions can't occur. EMI or cross-talk on the UTP cabling would cause corrupted frames and result in CRC errors. Oversize frame errors are Layer 1 errors. Duplicate MAC addresses would result in a Layer 2 error. Incorrect protocol or IP addressing configurations would result in Layer 3 errors.

REFERENCES

LabSim for Network Pro, Section 6.7.

Question 34: ✔ Correct

You need to configure the FastEthernet 0/1 interface on a switch to automatically detect the appropriate link speed and duplex setting by negotiating with the device connected to the other end of the link.

Drag the command on the left to the appropriate configuration step on the right. It is possible that not all of the commands are required.

Enter global configuration mode.Enter interface configuration mode.Set the speed of the interface.

✔ `conf t`

✔ `int fa0/1`

✔ `speed auto`

Set the duplex setting for the interface.

✔ `duplex auto`

EXPLANATION

To complete the requirements of this scenario, you need to use the following commands:

- Enter global configuration mode: **conf t**
- Enter interface configuration mode: **int fa0/1**
- Configure the interface to automatically detect the line speed: **speed auto**
- Configure the interface to automatically detect duplex settings: **duplex auto**

REFERENCES

LabSim for Network Pro, Section 6.3.

Question 35: ✔ Correct

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You completed the lab correctly.

View Lab Report

Question 36: ✔ Correct

A switch receives a frame addressed to the MAC address FF:FF:FF:FF:FF:FF.

What will the switch do with the frame?

- Forward it out the port and to the device with that MAC address.
- Process the frame and respond to the sending device.
- ➡ Send it out all ports except for the port it was received on.
- Drop the frame.

EXPLANATION

The MAC address FF:FF:FF:FF:FF:FF identifies the broadcast address, meaning all hosts on the subnet. Switches forward broadcast frames out all ports except for the port on which the frame was received.

REFERENCES

LabSim for Network Pro, Section 6.3.

Question 37: ✔ Correct

Which of the following best describes the concept of a VLAN?

- Devices in separate networks (which means they have different network addresses) logically grouped as if they were in the same network.
- Devices connected by a transmission medium other than cable (such as microwave or radio transmissions).
- Devices connected through the internet that can communicate without using a network address.
- Devices on different networks that can receive multicast packets.
- ➡ Devices on the same network logically grouped as if they were on separate networks.

EXPLANATION

A VLAN is created by identifying a subset of devices on the same network and logically identifying them as if they were on separate networks. Think of a VLAN as a subdivision of a LAN.

REFERENCES

LabSim for Network Pro, Section 6.4.

Question 38: ✕ Incorrect

To answer this question, complete the lab using the information below.
You have already answered this question.
You are not allowed to view the lab again.

Launch Lab

You did not complete the lab correctly.

View Lab Report

Question 39: ✓ Correct

Which of the following connectivity hardware is used to create a VLAN?

- Bridge
- Router
- Switch
- Hub

EXPLANATION

Specialized switches are used to create virtual LANs. The switch must be capable of appending and reading VLAN IDs.

REFERENCES

LabSim for Network Pro, Section 6.4.

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